



Society of Petroleum Engineers

SPE-229768-MS

Value Optimization of Nigerian Gas Fields with Various Maturity Levels, Through Integrated Asset Modelling

M. Repina and M. Darde, TotalEnergies, Pau, FRA

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229768-MS](https://doi.org/10.2118/229768-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Several infill wells and a new undeveloped field are planned to be integrated into the production of four legacy gas fields, each at different maturity levels, sharing complex surface network facilities. This study demonstrates how a fit for purpose coupled Integrated Asset Model (IAM) enhances production by: prioritizing the drilling sequence, scheduling workovers, optimizing the compression strategy. The coupled model of this asset was built in the integration platform. It combines five 3-dimensional gridded reservoir models, managed by a commercial reservoir simulator, some tens of tank material balance reservoir models and a complex production surface network. The gas surface network model considers: i) about forty gas wells, with several levels of completion, that can be routed to either one of four gathering networks with different pressure levels; ii) three compressors; iii) associated gas and gas lift from nearby oil wells; iv) routing changes with time.

The integration platform manages the coupling, facilitating all data exchanges between the network simulator and the reservoir models. The integrated model contains two main workflows: Well management and Field production optimization. These workflows aim to replicate field operations logic (e.g., closing wells at liquid loading risk, prioritizing wells with lower WGR, adapting wells routing) and suggest actions to (re)activate wells, prioritizing gas or condensate production. The constructed integrated model identified the most impactful surface network constraints and proposed timely actions to enable production maximization, such as: upgrading medium pressure compressor capacity; selecting the optimal internal diameter for a new flowline; revising the drilling sequence; optimizing the well completion strategy.

The integrated asset model was validated against historical measurements, with 2023 gas production predictions aligning with measured production. Pressure loss estimates for main pipes in the surface network model matched measured losses between different manifolds.

The model confirmed the technical feasibility of producing incremental stakes from both infill wells in legacy fields and a new undeveloped field, considering shared surface facility constraints. Several optimization and debottlenecking options were proposed to unlock the asset's potential, including delaying the switch from high pressure to medium pressure manifold for some new wells; upgrading medium pressure compression capacity; revising the re-completion sequence.

Thanks to the high scalability of the reservoir simulator, and to the massive computation resources of our HPC cluster, simulating the 25-year forecast in the coupled integrated model takes 12 hours, allowing the model to be used for regular optimization.

Introduction

In the complex and dynamic world of asset management, the significance of Integrated Asset Modeling (IAM) cannot be overstated. IAM serves as a holistic approach that unifies various components of asset development, from the reservoir to the processing facilities, enabling a flow of information and decision-making processes. It offers a deeper understanding of the interactions between different asset components, which is crucial for effective management and optimization. Moreover, IAM breaks down silos between disciplines, fostering collaboration and integration, which is essential for maximizing asset value and ensuring operational efficiency.

During the second quarter of 2023, TotalEnergies OML58 asset (developed in partnership with NNPC, TotalEnergies having 40%), requested a review of the current IAM. This analysis is crucial for the Ubeta development and the Obagi Deep/Ibewa Infills projects, both of which are targeting Final Investment Decisions (FIDs) in 2024.

Stand-alone production profiles and reserves were computed for each individual project and validated by an appropriate committee. Nevertheless, following the technical committee recommendation, an integrated asset model was requested to consider surface constraints and capacity limits. The primary goal of this study was to verify the available ullage to produce the incremental shares of both projects, while ensuring optimal production efficiency.

Scope of study

OML58 is situated approximately 80 kilometers northwest of Port Harcourt in the central Niger Delta, Nigeria, as shown in Fig. 1. It comprises three mature oil fields—Obagi, Erema, and Olo—and three gas fields—Ibewa, Obagi Deep, and Obagi Gas Cap—along with an undeveloped gas field, Ubeta. The gas produced from OML58 is exported to the Nigerian LNG plant on Bonny Island and also supplies the domestic market.

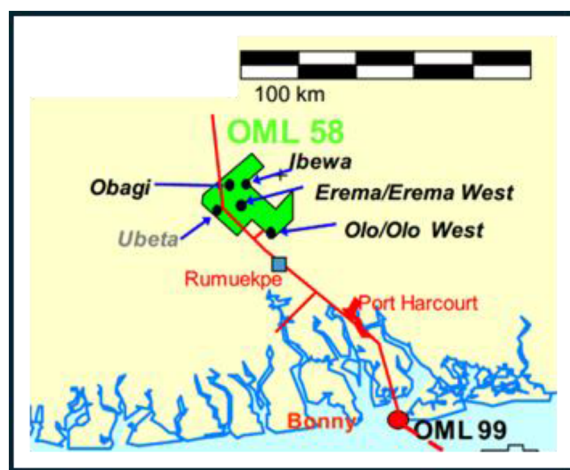


Figure 1—Figure OML58 Mining acreage

In 2024, two Final Investment Decisions were scheduled: the first for drilling infill wells in the Ibewa field and the second for developing the Ubeta green gas field. The planned production from these Ibewa

infill wells and the undeveloped Ubeta field will enhance the output from three existing gas fields of varying maturity, which share common surface network facilities.

The IAM model was originally constructed in 2004 during the Pre-project phase. In 2023, it became evident that the surface network model was outdated, requiring a thorough revision of both the wells and the surface facilities. Additionally, it was mandatory to recode the Well management and Field Target Management operational logics to ensure they accurately represented the on-site reality.

As a result, the goals of this study have been established as follows:

- Conduct a thorough review of the Surface network model
- Validate the production network (dP pipes predicted vs. measured)
- Rewrite the Well management and Field Target Management logic to align with the present operational strategies
- Incorporate the existing 3D Ibewa reservoir models into IAM
- Add the new Ubeta project into IAM
- Refresh all reservoir models according to the most recent history match outcomes
- Identify the limiting constraints
- Recommend timely measures to enable the maximization of production
- Serve as a primary instrument for estimating reserves and resources

Data needed

To construct the integrated model, the following data were required:

- History Matched 3D Reservoir Models for several Ibewa layers
- Project 3D Reservoir model for Ubeta field
- Up to date material balance tank models for all other fields / layers
- The Network schematic: all present surface facilities, wells and flowlines connecting them to the manifolds.
- Validated to recent well tests well models for each well or generic models for new wells
- The Network constraints: Compression capacities, Chokes, Separator Pressure, Rates
- The topology of the pipes (ID, length).
- Downtime of each part of the model: network and wells or global value
- The flow assurance study report and all input (erosion mixture velocity, turndown rate).
- Well constraints: Max Drawdown, Min Qgas, Max WGR
- Other input: Associated gas coming from oil wells, Gas lift, Fuel gas, Water injection rate in Obagi tanks.

Models and Assumptions

Reservoir models

The Integrated Asset Model (IAM) includes four 3D reservoir models, which cover three Ibewa layers and the Ubeta field. Additionally, it incorporates several tens of material balance tanks that are directly connected to the Surface network model. These material balance tanks represent different layers of the Obagi Gas Cap,

Obagi Deep, and Ibewa fields. Structural top reservoir map is shown in Fig.2. All reservoir models for fields and layers in production have been matched to historical data up to January 2023. The study considers a total of 40 wells, with several perforations in the Obagi Deep, Ibewa, and one Ubeta well that might be activated sequentially.

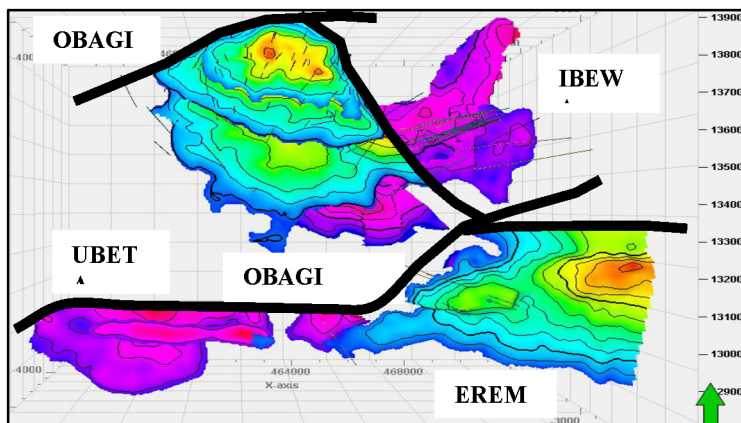


Figure 2—Structural top reservoir - view from top

Well models

Well models were aligned with recent production tests for producing wells, while generic models were used for new wells. The reservoir well datum was set at the depth of the last node of vertical list performance (VLP) curve as provided by the well model files. VLP curves were then generated from the validated well model files, ensuring that the ranges of Gas Rate, Top Node Pressure, Water-Gas Ratio (WGR), Gas-Oil Ratio (GOR), and Condensate-Gas Ratio (CGR) fully covered the forecaster's operational envelope. Additionally, the spacing was adjusted to ensure a smooth shape of the VLPs. All VLPs are generated in black oil model.

For the wells connected to material balance tanks, the inflow performance section was pulled from the corresponding wells models, as shown in Fig. 3. In the IAM, all wells were set to be controllable. The DP choke was used as a control mechanism to optimize production while ensuring compliance with all constraints.

IPR Type	Forchheimer	Layer Pressure	2451.1411	psia
Darcy	515.99136	psi ² /(Mscf/day)	Layer Temp	65.000002
Non Darcy	0.00775309	psi ² /(Mscf/day) ²		deg C
IPR dP shift	0	psi		

Fluid Properties				
Cond. gravity	44.692253	API	Water salinity	10000
				ppm
Gas gravity	0.607938	sp. gravity	H ₂ S	0
				percent
CGR	0.426896	STB/MMscf	CO ₂	3.02
				percent
WGR	0.30753	STB/MMscf	N ₂	1
				percent
☐ Use tank impurities				

Figure 3—Example of Inflow Performance characteristics (IPR type, value and fluid properties) for one Obagi Gas Cap well

Pipe description

All flowlines were thoroughly checked and updated with the most accurate characteristics, including bathymetry, length, and internal diameter. Following internal flow assurance recommendations, the OLGA 2P correlation was used in the IAM for all main flowlines, with an additional 10% margin applied to frictional and gravitational pressure losses. Additionally, a maximum mixture velocity constraint of 20 m/s was set for the Ibewa MP pipe.

The Rough Approximation option was activated in Surface Network model for temperature modeling. To evaluate the impact of condensed water vapor on pressure losses, the corresponding option was enabled. Prior to this, the reservoir temperature for each well was updated.

PVT

The description of the PVT models in the IAM is as follows:

- All well VLPs are in the black oil model with a rough approximation of the temperature model.
- Reservoir models in the Black Oil Tables
- Surface network is modeled using Black Oil correlations, as illustrated in Fig. 4.



Figure 4—PVT black oil correlations used in the surface network model

Surface network modelling

The OML58 surface installations encompass the OGBOGU Flow Station (OFS), where all compressors are situated, the Ibewa cluster manifold, and the OBITE Treatment Centre (Gas plant), as depicted in Fig. 5.

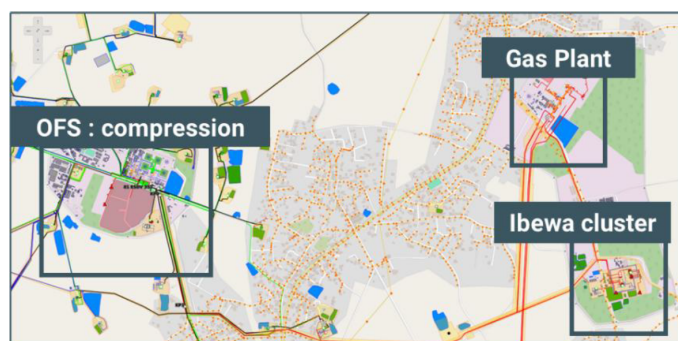


Figure 5—Helicopter view of Surface facilities of OML58

The gas compression system consists of four separators. Initially, gas at the VLP node is compressed from 4 bara to the LP stage at 9 bara. It is then further compressed to the MP stage at 30 bara, and ultimately to the HP stage at 80 bara. A simplified diagram of the Obagi gas compression system is depicted in Fig. 6.

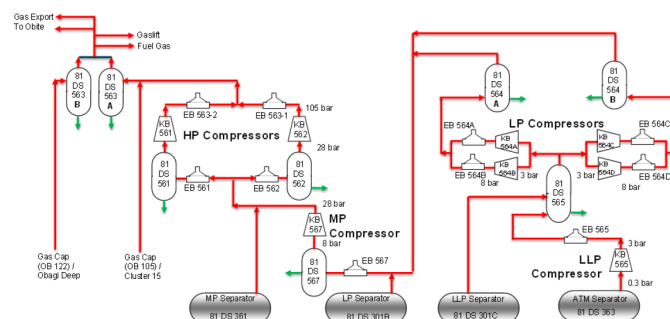


Figure 6—Gas compression system at OBAGI Flow Station

The compression system is represented in production surface network model using separators and source objects. The arrival pressure at each separator is predetermined and set within the input table in the integrated platform. An illustration of the VLP to LP compressor can be found in Fig.7.

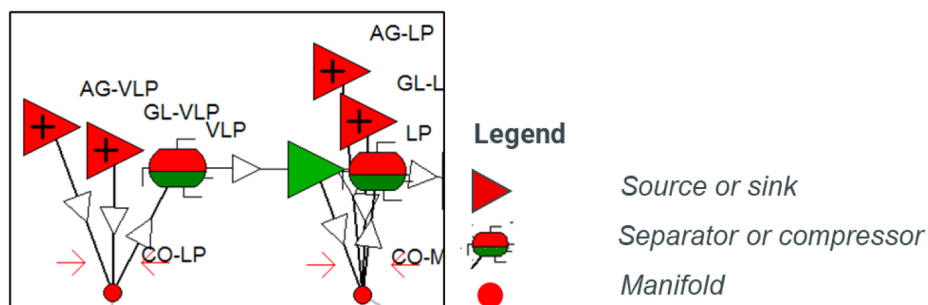


Figure 7—Simplified way of modeling gas compression in the Surface Network Model. AG- associated gas, GL- gas lift, LP- low pressure, VLP – very low pressure.

The full gas production surface network is shown in Fig. 8.

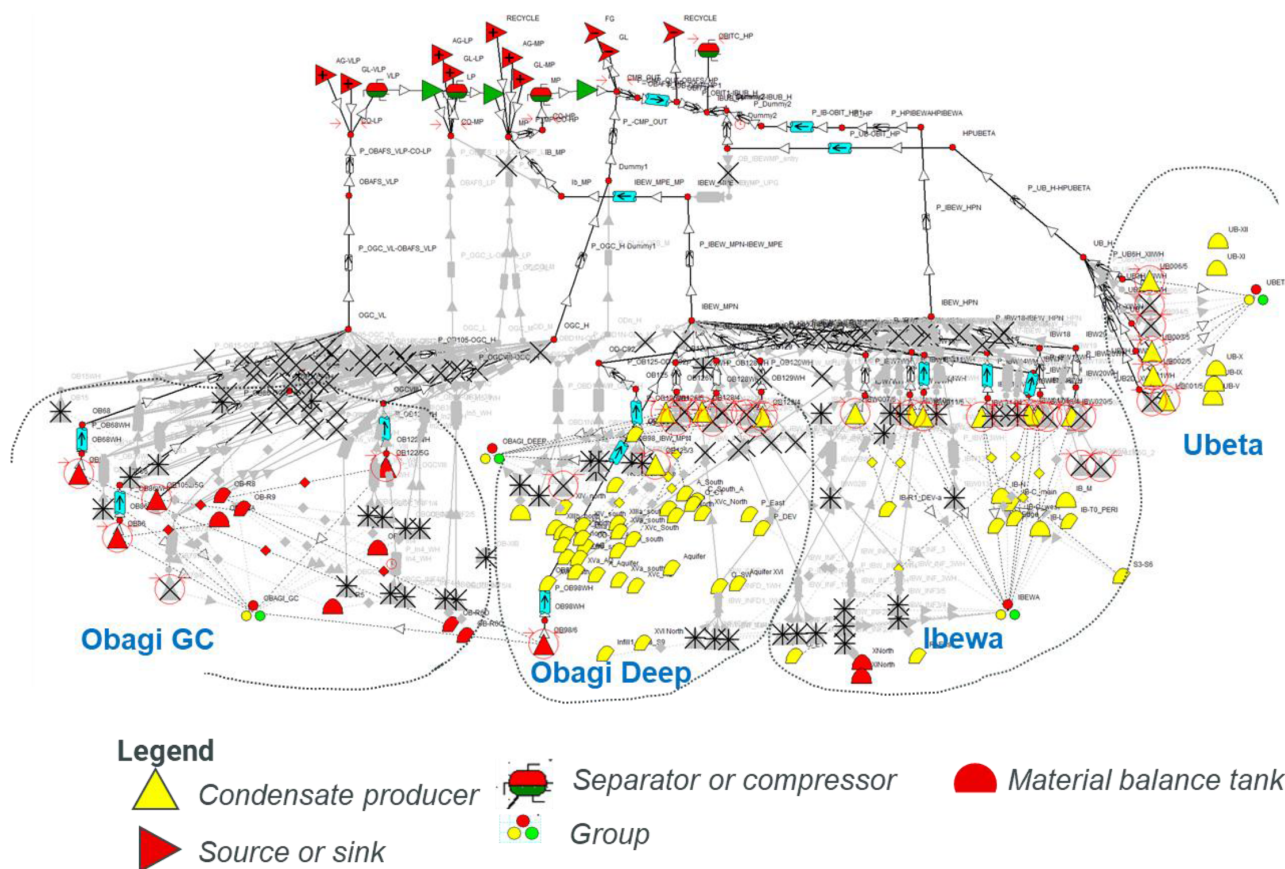


Figure 8—Production surface network of OML58 asset

Well constraints

The constraints specified through one of the visual workflows include:

- Maximum Drawdown
- Maximum Gas rate

- Turner Rate
- Minimum Gas Rate
- Maximum Water-Gas Ratio (WGR)
- Maximum Cumulative Gas
- Downtime

These constraints come from the input table established in the integration platform, except for the Turner Rate, which is computed by the workflow. At the beginning of each timestep, they are automatically transferred to the surface network model.

The well opening dates, information about the associated re-completions and available connections to manifolds are informed in the second table in the integrated asset model.

Process and field management input and constraints

All input and constraint values provided in the surface network are instructed in one of the tables in the integrated asset model. This table includes:

- Total Associated Gas sourced from adjacent Obagi oil wells
- Distribution of AG across VLP/LP/MP pressure nodes
- Total Gas Lift (GL) quantity
- Allocation of GL among VLP/LP/MP pressure nodes
- Global Uptime Factor
- Fuel Gas consumption
- OFS Flare Limit
- Capacities of MP, LP, and VLP Separators
- Target at gas plant (constraint for gas treatment at the primary separator)

Integration platform

Integrated asset model of OML58 consists of:

- Production surface network for all fields.
- 3D reservoir models representing Ibewa level C, Ibewa level O, Ibewa level P, Ibewa level R, and Ubeta field.
- Several tens of material balance tank models for the remaining reservoirs/layers, which are directly integrated into surface network model and are not coupled with the integration platform.

[Fig.9](#) illustrates the layout of the OML58 integrated asset model.

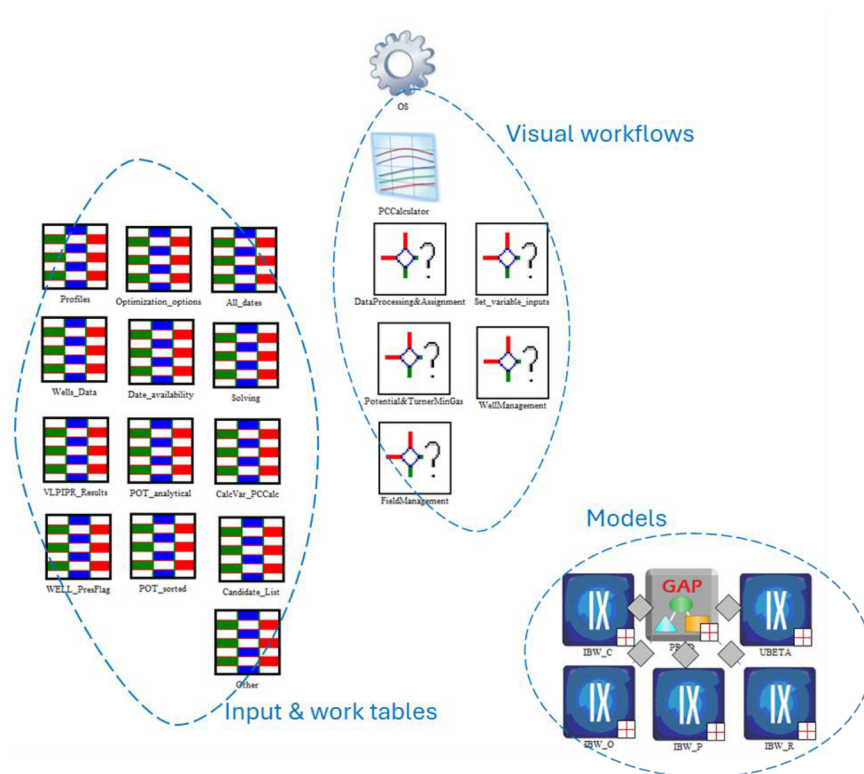


Figure 9—Main view of OML58 Integrated asset model

General

The integration platform manages the coupling and facilitates all data exchanges between the network simulator and the reservoir models. The numerical scheme for the applied coupling is referred to as explicit, as communication between the reservoir and surface models occurs only at fixed intervals, with each application acting independently in between. The integrated platform establishes the connection between the reservoir model and the surface network model, allowing the PVT and inflow (IPR) properties to be passed to the surface network solver. Once the optimum rates are determined by the surface network solver, they are passed back to the 3D reservoir model. This sequence is then repeated, with no further synchronization to check the consistency of production surface network and 3D reservoir model results. Quality control is left to the engineer, who compares the bottom hole pressure (BHP) and rates from production surface network and 3D reservoir models, as shown in the Coupling consistency section.

Workflows

The integrated model includes several logical scenarios implemented through visual workflows, which are as follows:

- Input & constraints assignment: A) pre-treat the data and set constant inputs that do not vary with time. Example: initialize production surface network model schedules, set water injection schedule, initialize well connections at the beginning of the forecast. B) Set inputs that varies with time. The instructions are set at the beginning of each time step. Example of actions treated by this workflow are: assignment of GL, AG, gas plant target, compression capacities, well opening etc.
- Analytical well potential calculation: utilizing the Performance Curve (PC) Calculator of the Integrated asset model, this workflow provides the calculation of well rates across a variety of associated wellhead pressures (WHP).

- Turner rate calculation: Using the same integrated object from PC Calculator, this workflow computes the Turner velocity and Mixture velocity for the wells, considering various WHPs. The results from this computation help identify instances where wells may face liquid loading issues.
- Well management workflow: executed at each time step to monitor wells for any limiting constraints, such as WGR or minimum gas rate, or if closure instructions have been issued. If a well meets any of these conditions, the workflow assesses whether the well can be reactivated at a lower pressure node. If reactivation is not feasible, it then evaluates the possibility of opening a subsequent recompletion. A very simplified flow chart emulating the real Well management workflow is demonstrated in Fig. 10. The actual well management workflow as programmed in the IAM is illustrated in Appendix 1.
- Field management (gas plant target meeting): executed at each time step to monitor if the target set at the gas plant is met or not. When it is not feasible to individually transition wells to the lower pressure node, such as with Ubeta, the workflow calculates the consequences of putting all associated wells collectively to the lower pressure node. A very simplified flow chart emulating the real Field management workflow is demonstrated in Fig. 11. The actual field management workflow as programmed in the IAM is illustrated in Appendix 1.

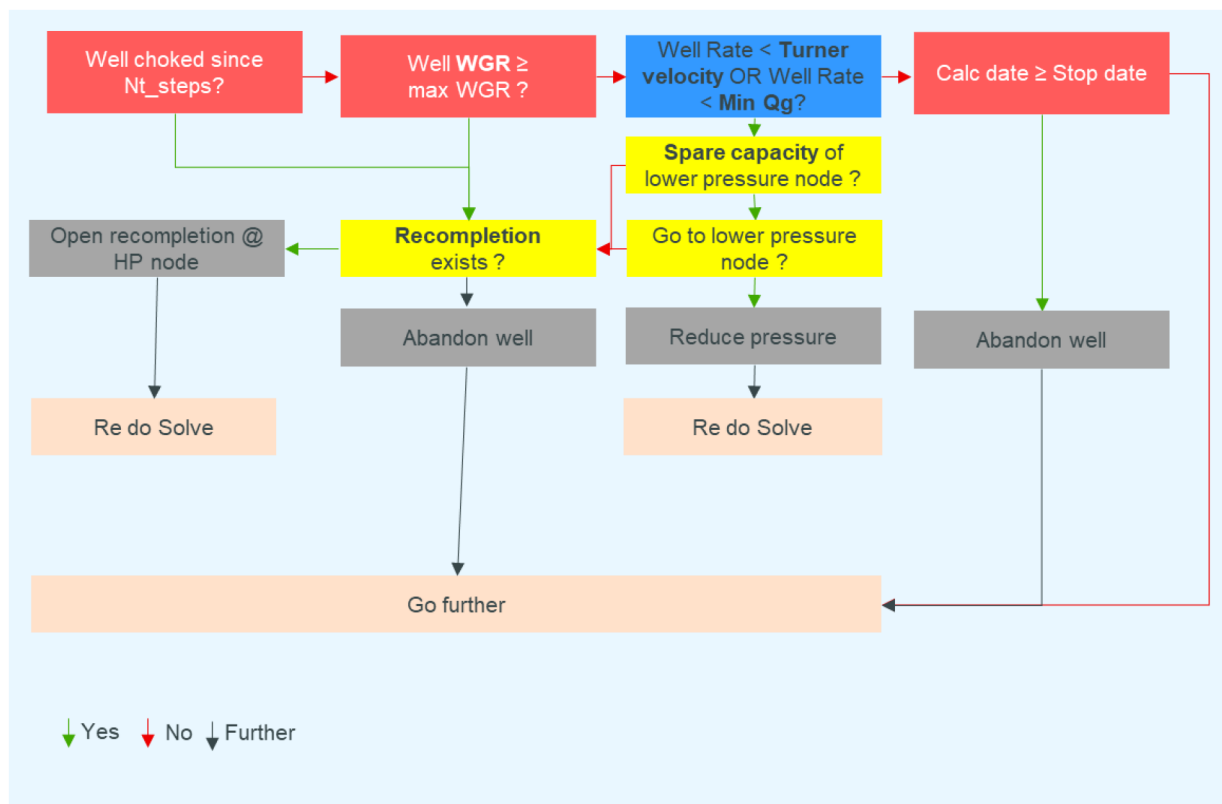


Figure 10—Simplified demonstration of Well Management workflow.

Note: @ beginning of each t, from Prosper models: AnalyticalPotential (POT) for different pressure nodes

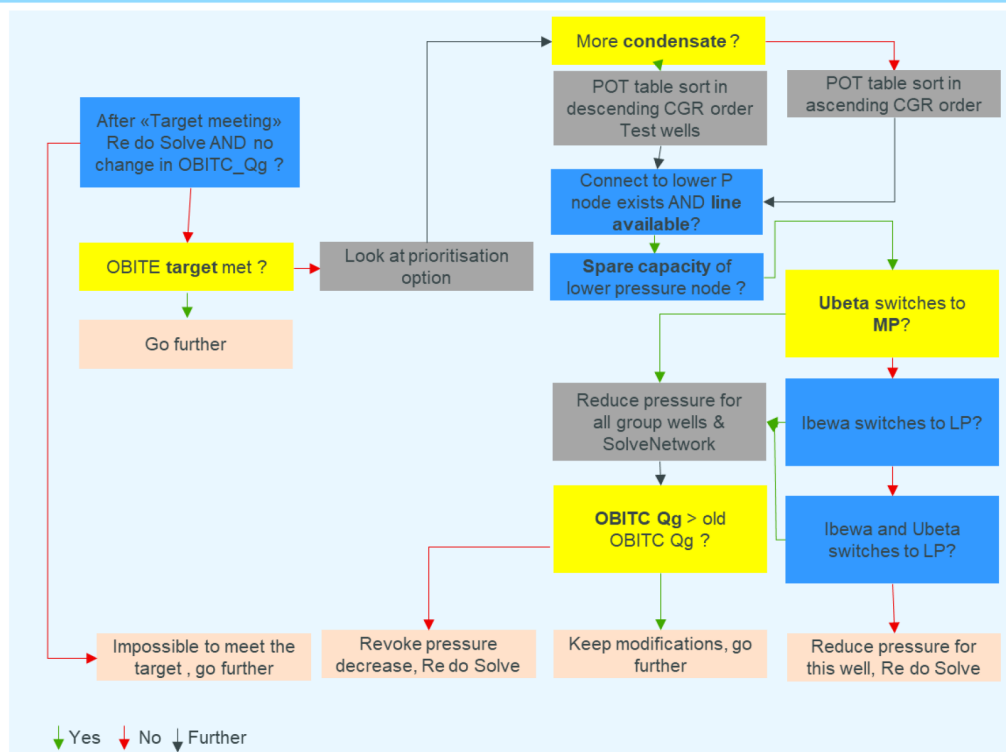


Figure 11—Simplified demonstration of Field Management workflow. OBITC_Qg refers to Gas rate at gas plant.

Results

Coupling consistency

As mentioned above, comparing the BHP and rates from production surface network model and 3D reservoir models is essential to ensure the quality of the coupling. Fig. 12 illustrates this analysis for two wells in the Ubeta field. As shown, the gas rate, liquid rate, cumulative gas rate, and BHP are closely aligned, indicating that the IPR block scaling and explicit coupling are of good quality. The absolute difference in cumulative gas rate is under 1%.

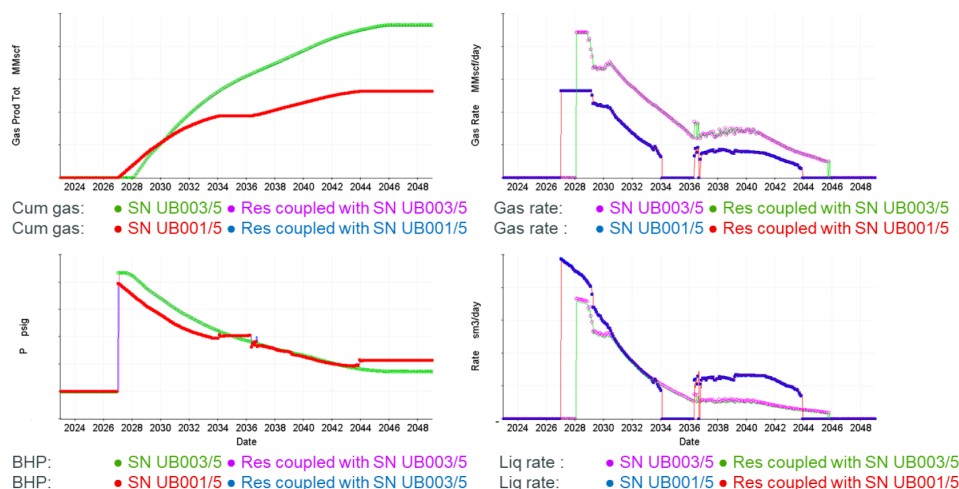


Figure 12—Coupling QC for two Ubeta wells. Where SN – surface network, Res – 3D reservoir model.

Field management consistency

Wells producing with high Water-Gas Ratios (WGR) or below the minimum gas rate (Turner equivalent rate) are targeted for remediation. This process may involve abandoning the current completion (if WGR is high) and opening the next available one, or attempting to place the well on a lower pressure node (if gas rates are low) and there is spare compression capacity.

Fig. 13 shows two examples of consecutive completion opening for OB128 and OBD1N wells.



Figure 13—Demonstration of completion piloting with Well Management logic of an IAM. OB128 well has 3 completions, OB128/5 is closed with the stop date, OB128/4 is opened with the start date, OB128/3 is opened by the workflow when the previous one is flooded. OBD1N well has 2 completions, OBD1N/3 is opened after the closure of the previous one because of high WGR.

Example of consecutive switch to the lower pressure node is demonstrated in Fig. 14.

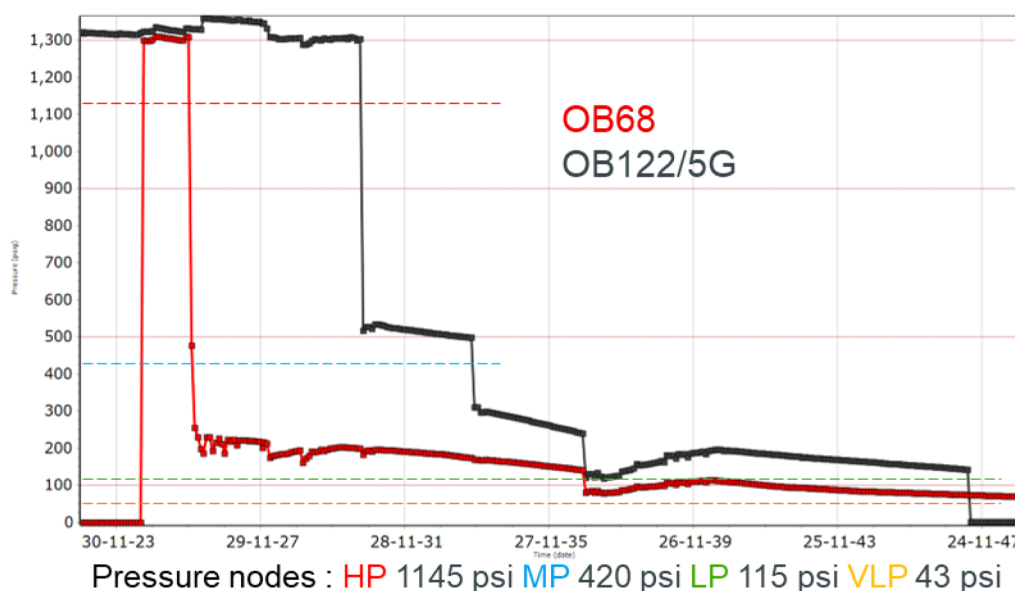


Figure 14—Demonstration of well reactivation by lowering the pressure node. OB68 goes over all possible manifold pressures, while OB122/5G is stopped at LP.

Fig. 15 confirms that the Field Management workflow is functioning as anticipated. Specifically, in this run, the Ubeta Medium Pressure (MP) pipe is planned to become operational in January 2036. However, the transition of all Ubeta wells to the MP model is realized a few months later. This delay is due to the immediate saturation of the MP separator upon the simultaneous shift of all Ubeta wells, which would lead to the choking of the baseline wells. Therefore, the workflow strategically searches for the most suitable date to optimize production on a global scale.

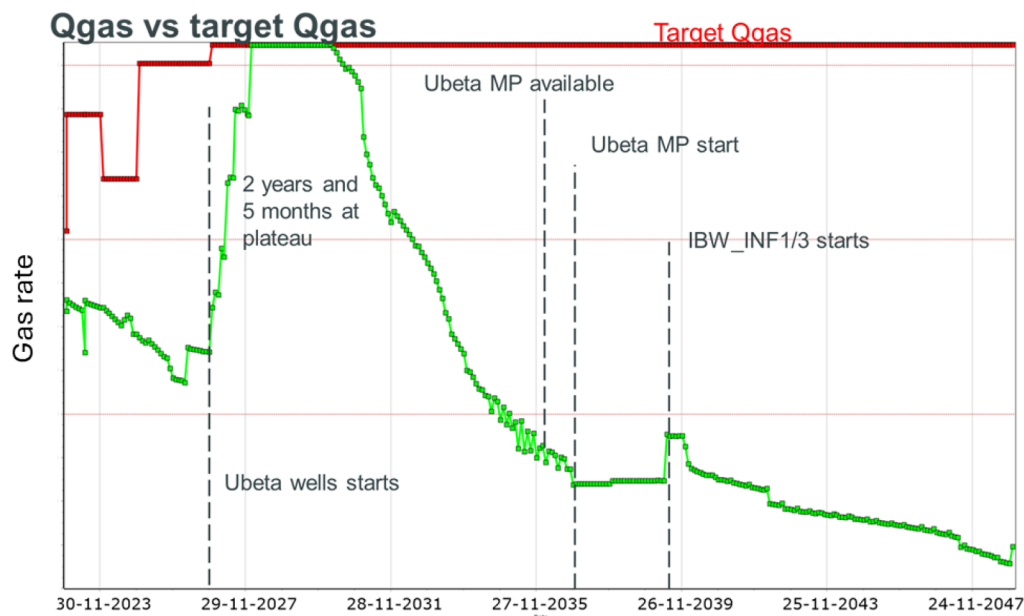


Figure 15—Gas Rate profile at OBITC gas plant

Flow Assurance consistency

To affirm that surface network model accurately predicts pressure losses, a validation exercise was conducted. This involved a comparison of the estimated pressure losses in three primary pipelines—Ibewa_MP, Ibewa_HP, and OFS_OGP—as calculated by the surface network model, against the actual measured values.

The validation was achieved by allocating the specified flow rates to the fluid sources at each tested manifold and then determining the pressure drop across the respective pipe. The systems configured to validate the pressure differentials of the three pipes are illustrated in Fig.16 A-B-C. For the major pipes, the difference between the measured and model-predicted pressure drops (DP) is less than 10%.

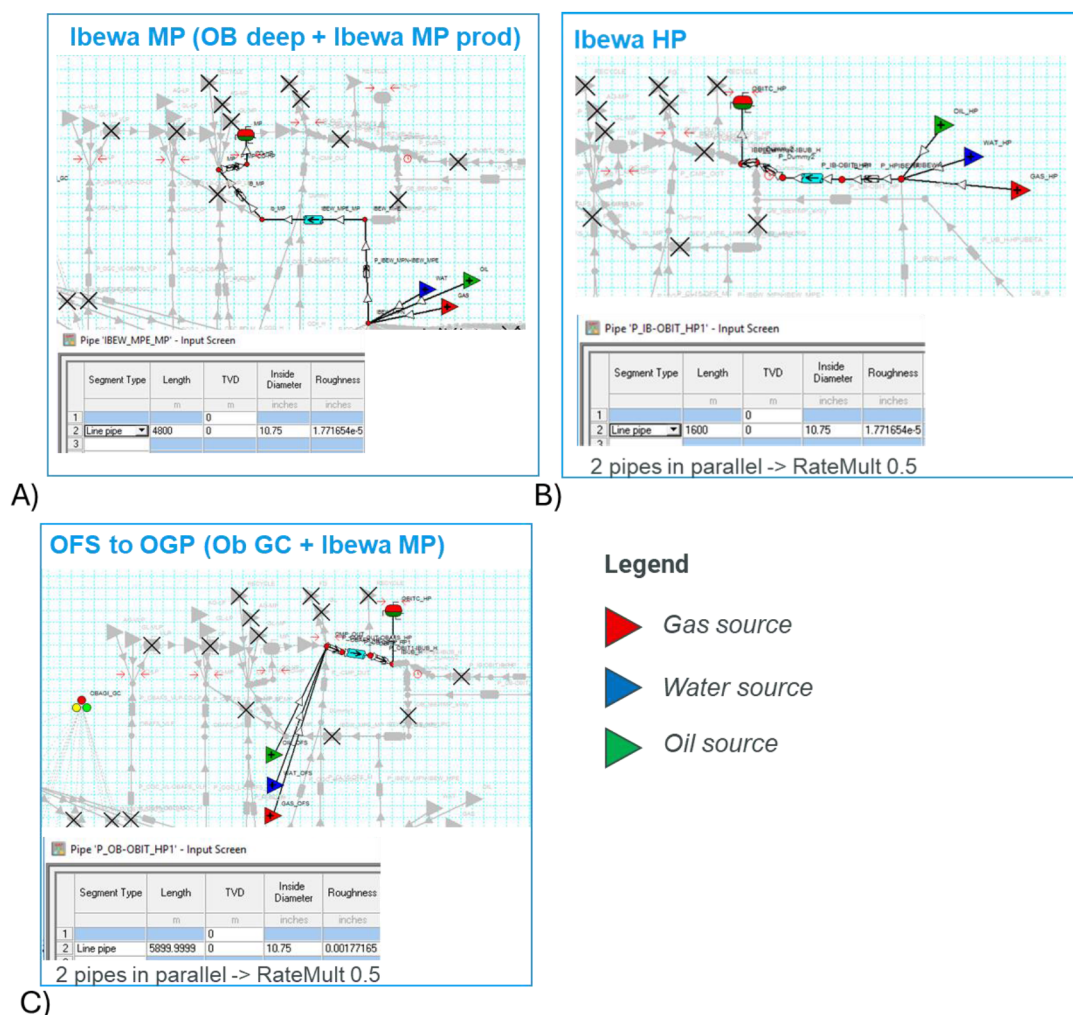


Figure 16—Surface network sub-models used to validate pressure drops of the main flowlines of OML58

An example of the comparison between the measured and predicted pressure drops for the Ibewa_MP pipe is shown in Fig. 17.

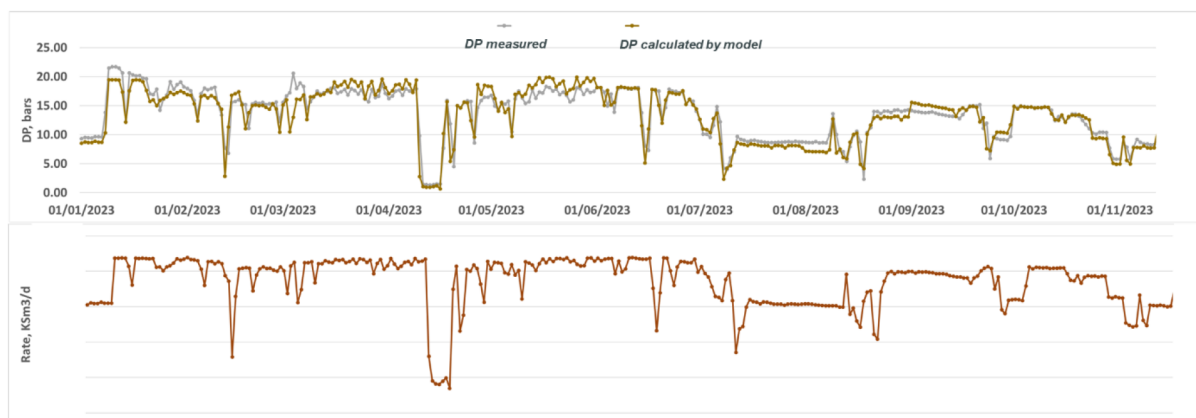


Figure 17—The comparison of measured and model-predicted pressure losses in the Ibewa_MP pipe, along with the associated gas rates used for this calculation.

During the postmortem analysis, the evolution of the maximum mixture velocity in the Ibewa_MP pipe was examined. It was found that the erosion velocity limit of 20 m/s, as specified by Flow Assurance, was

met on two occasions. Fig. 18 illustrates the profiles of maximum mixture velocity. Further analysis of the gas rate profile for the IBW_MP pipe indicates a risk of exceeding the erosion threshold of 20 m/s at a Gas Rate of 2.5 Msm³/d.

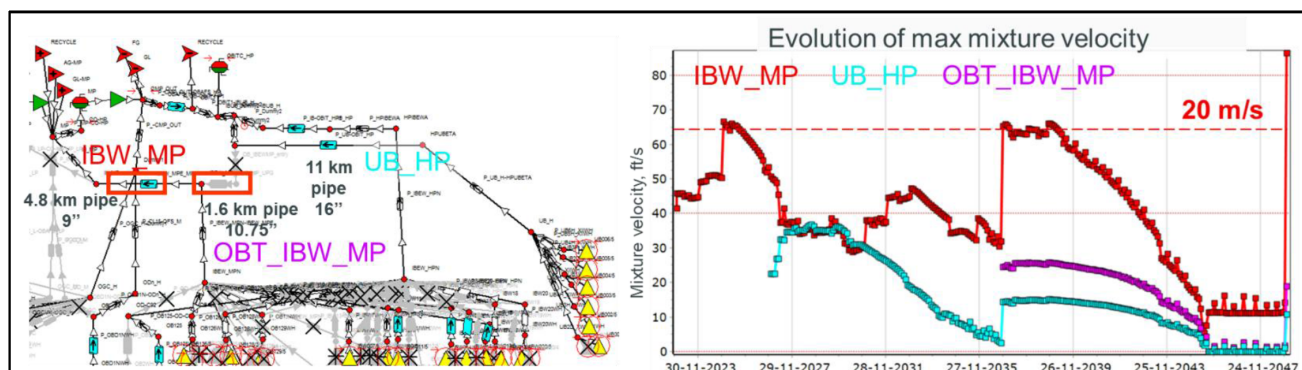


Figure 18—Evolution of maximum mixture velocity in Ibewa_MP, OBT_Ibewa_MP and Ubeta_HP flowlines.

Production consistency

The IAM was first run with current wells, starting on 1st Jan 2023, to confirm that it globally represents the current potential of the OML58 in 2023, as shown in Fig. 18. Minor divergences are attributed to the fact that actual downtime was not considered in the model. From Fig. 19 it is seen that IAM results are in line with the real production measured from Jan 2023 to Jan 2024.

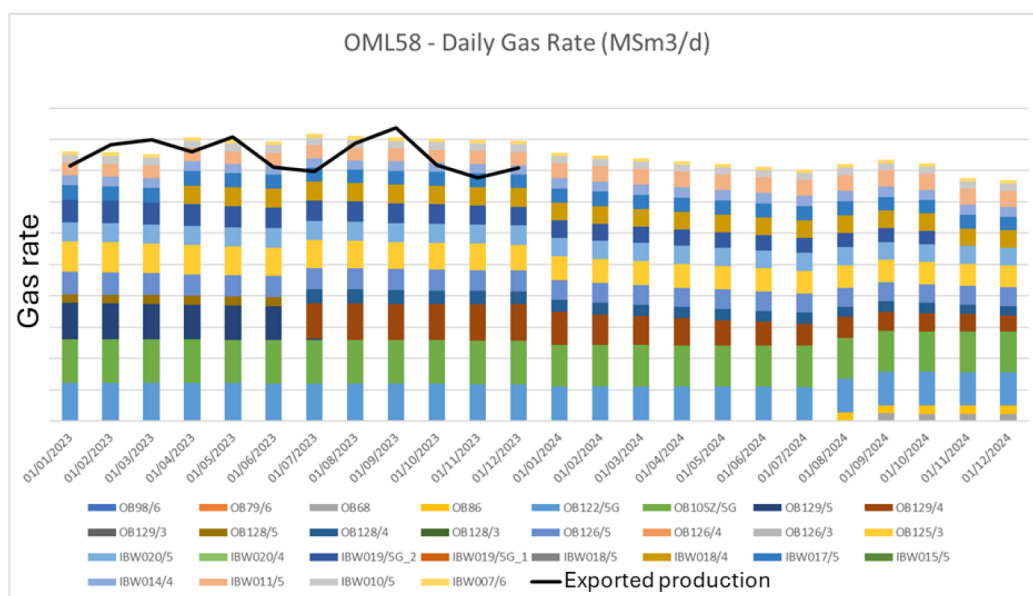


Figure 19—Profiles of gas rate predicted by IAM vs the actual production.

Analysis of compression capacities

The results presented in this section pertain to the run that includes Baseline wells, Infill wells, and Ubeta wells in both HP and MP production.

The primary limitations affecting the production profiles include:

- Obite Plant Capacity: This capacity acts as the principal constraint on OML58's production capability, particularly when Ubeta starts the production.

- **MP Compressor Capacity:** This constraint is frequently met by the Ibewa, Obagi, and Obagi Deep wells, as well as when Ubeta is switched to MP pressure node.

The profiles of Gas Rate at MP, LP and VLP compressors are shown in Fig.20.

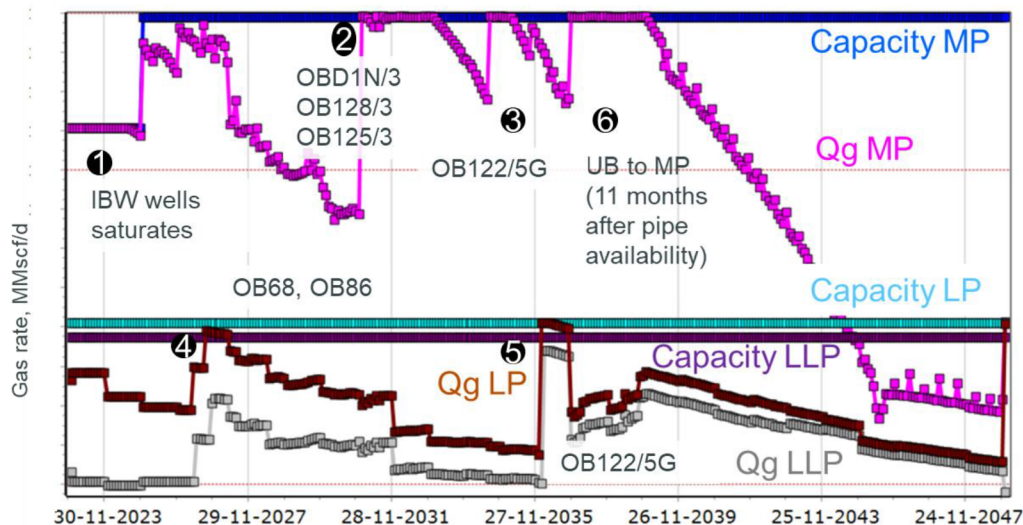


Figure 20—Profiles of gas compression (capacity vs production): 1. Saturation of MP compressor with Ibewa wells; 2. OBD1N/3, OB128/3 and OB125/3 connected to MP manifold; 3. OB122/5G got connected to MP manifold; 4. OB68 and OB86 nearly saturate the LP compressor; 5. OB122/5G switched to LP manifold and saturates the LP compressor; 6. The Ubeta wells have been connected to the MP manifold, resulting in the saturation of the MP compressor.

Incremental production from new wells

Infill wells. The Ibewa and Obagi Deep Infills project encompasses three wells:

- 1st Ibewa infill: targeting the deepest layers of Ibewa, specifically levels R, S, and T, with an anticipated production start in late 2025. It is important to note that level R is contingent and is thus excluded from this analysis.
- 2nd Ibewa: targeting levels C, O, and P within Ibewa, this well is expected to start producing in mid-2026. Note that level P is contingent and is not considered in this study.
- Obagi Deep: This well targets the unproduced levels XI and XIV of the Obagi Deep field, with a projected start being mid-2026.

All three wells are equipped with selective completion systems, enabling sequential activation of the different levels (no commingle production).

The production forecast analysis results demonstrated that adding these three wells has a very limited impact on the baseline, primarily towards the end of the forecast period.

New development wells. The Ubeta field is planned to be developed with six wells targeting levels V, IX, X, XI, and XII. First gas production is expected in 2027 with three wells connected initially, followed by the remaining three wells being connected a few months later.

The results presented below take into account that Ubeta wells operate in both HP and MP modes.

Since all Ubeta wells must switch to MP simultaneously (due to a single export line), the IAM model determines the optimal date for this switch to maximize total production. Although the export line becomes available in Jan. 2036 (input given in the model), the model strategically schedules the switch to the MP manifold several months later to optimize global production while considering the MP compressor's

capacity. As shown in Fig.21, for approximately two years following the switch, Ubeta's production is capped by the MP compressor capacity.

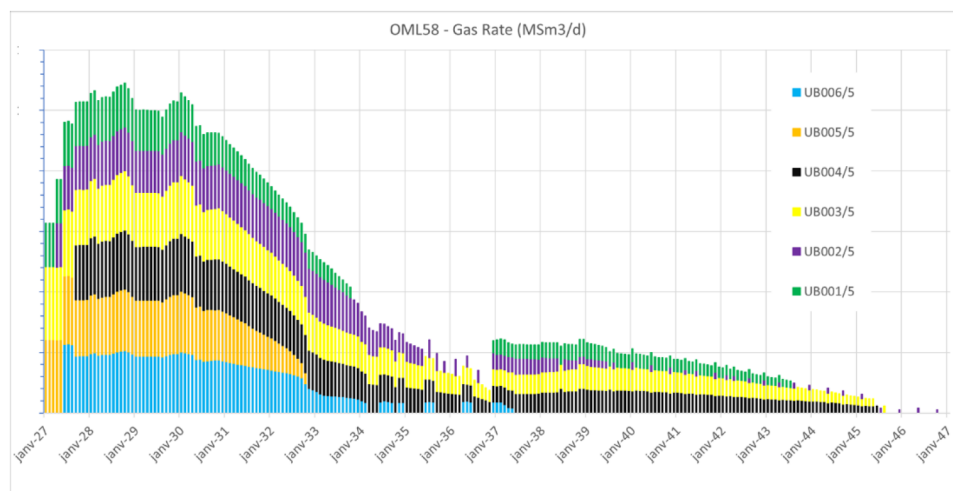


Figure 21—Gas production profiles for Ubeta wells.

With priority given to the higher potential Ubeta wells, the baseline wells are choked or shut-in during this period. Once Ubeta production declines from its plateau, these baseline wells can be gradually un-choked. However, the activation of the Ubeta wells at the MP pressure node has an additional negative impact on the baseline. Despite this, the incremental resources provided by the MP are significant.

Conclusions

A reliable and robust Integrated Asset Model has been successfully developed for the Nigerian OML58 asset. By integrating a surface network model, multiple 3D reservoir models, and numerous material balance models, all guided by comprehensive logics representing well and field management operational philosophies, a consistent integrated model has been created. This model encompasses 40 wells, several reservoirs, 3 compressors, and a production separator.

The final simulations performed with the IAM led to the following observations:

- The model confirmed the technical feasibility of producing the incremental stakes of both projects (infill wells in legacy fields and a new undeveloped Ubeta field) while considering the constraints of the shared surface facilities.
- The IAM helps identify the most impactful surface network constraints and propose timely actions (e.g., upgrading the medium pressure compressor capacity, determining the optimal date for switching to the lower pressure node and opening new completions, revising the drilling sequence, etc.), thereby enabling production maximization.
- The IAM validated a technical addition of 2P reserves.
- The production shortage factor (uptime) is one of the most uncertain parameters that could significantly impact gas production.
- The compression capacity of the MP system is the most limiting constraint. Prioritizing the upgrade of the mid-pressure compressor's capacity is essential.
- At the individual well level, the primary cause for shutting in wells is the arrival of water, indicated by a high Water-Gas Ratio. Only two wells have been shut down due to issues with liquid loading.

Well management workflow as programmed in the integration platform.

